



Heart Disease in Ontario, 2010–2016

How we cleaned the data, what we did with it, and what it shows

Ontario Heart Monitoring Analytics (OHMA) · Group 5

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Big Data 2 · Built in R (RStudio)

The data at a glance

Source: Public Health Ontario · Ontario adults (20+), heart disease cases and rates, one row per year × sex × age group

7

years

2010 → 2016

126

rows

7 yrs × 3 sex groups × 6 age groups

6

columns

Age group, Sex, Year, Rate per 100k, Cases, Population

Key idea: a rate per 100,000 lets us compare groups of very different sizes fairly.

Sample rows (2010)

Age	Sex	Rate /100k	Cases	Population
20+	Both sexes	224	23,470	10,492,970
35–49	Both sexes	63	1,970	3,144,930
80+	Both sexes	1,300	7,320	563,080
20+	Females	168	9,110	5,408,000
20+	Males	282	14,360	5,084,970

“20+” and “Both sexes” rows are totals of the other rows — important for cleaning (next slide).

Step 1 — Cleaning the data

Four checks before any number was trusted



Fix formats

Age groups arrived wrapped in quote marks ('20+') and counts had thousands separators (23,470). Stripped both so R reads them as clean labels and numbers.



Tidy column names

R imported “Rate (per 100,000)” as X.Rate..per.100.000..
— renamed to Rate so every step reads clearly.



Avoid double counting

“Both sexes” = females + males, and “20+” = all age bands. Totals were kept only for totals, never added to their own parts.



Validate

Recomputed $\text{Cases} \div \text{Population} \times 100,000$ and compared with the published rate — they match (e.g. $23,470 \div 10.49\text{M} = 224$).

Step 2 — Processing and presenting

The same six moves, repeated for every question we asked



1. Import

Load the CSV into R



2. Filter

Keep only the rows a question needs (e.g. adults 20+, 2016)



3. Select

Keep only the columns needed



4. Add measures

New columns such as % of population affected



5. Compare

Men vs women, year vs year, age vs age



6. Chart

Turn each comparison into one clear graph

Tools: R with dplyr (filter, select, arrange, mutate, rank) for steps 2–5 and ggplot2 for charts. Earlier we also practised building the data by hand as vectors, matrices and data frames.

Checking the raw data

In R

Our RStudio session: inspecting columns and confirming nothing is missing

```
6 mydata2
7 # Showing data without Year and Population columns
8 mydataminus = select(mydata, -Year, -Population)
9 mydataminus
10 # Selecting variables contain 'X' in their columns names
11 mydata %>% select(where(function(x) grepl("X", colnames(x))))
12
```

12:40 (Top Level) ⌵

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Semester 3/big data 2/

```
126 2016 249310
> mydataminus = select(mydata, -Year, -Population)
> mydataminus
```

	Age.Group	Sex	X.Rate..per.100.000..	Number
1	'20+' Both sexes		224	23470
2	'20-34' Both sexes		4	110
3	'35-49' Both sexes		63	1970
4	'50-64' Both sexes		246	6660
5	'65-79' Both sexes		553	7420
6	'80+' Both sexes		1300	7320
7	'20+' Females		168	9110
8	'20-34' Females		1	20
9	'35-49' Females		28	450
10	'50-64' Females		123	1710
11	'65-79' Females		400	2880
12	'80+' Females		1149	4040
13	'20+' Males		282	14360
14	'20-34' Males		6	80

select() — look at the columns

Exposed the problems we then fixed: quote marks around age groups ('20+') and the garbled rate column name X.Rate..per.100.000..

```
19
20 #Below we are using two variables Sex and Year to determine uniqueness
21 x5 = distinct(mydata, Sex, Year, keep_all= TRUE)
22
23
24
```

20:7 (Top Level) ⌵

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Semester 3/big data 2/

```
4 2013 TRUE
5 2014 TRUE
5 2015 TRUE
7 2016 TRUE
> x5 = distinct(mydata, Sex, Year, keep_all= TRUE)
> x5
```

	Sex	Year	keep_all
1	Both sexes	2010	TRUE
2	Females	2010	TRUE
3	Males	2010	TRUE
4	Both sexes	2011	TRUE
5	Females	2011	TRUE
6	Males	2011	TRUE
7	Both sexes	2012	TRUE
8	Females	2012	TRUE
9	Males	2012	TRUE
10	Both sexes	2013	TRUE
11	Females	2013	TRUE
12	Males	2013	TRUE
13	Both sexes	2014	TRUE
14	Females	2014	TRUE

distinct() — is anything missing?

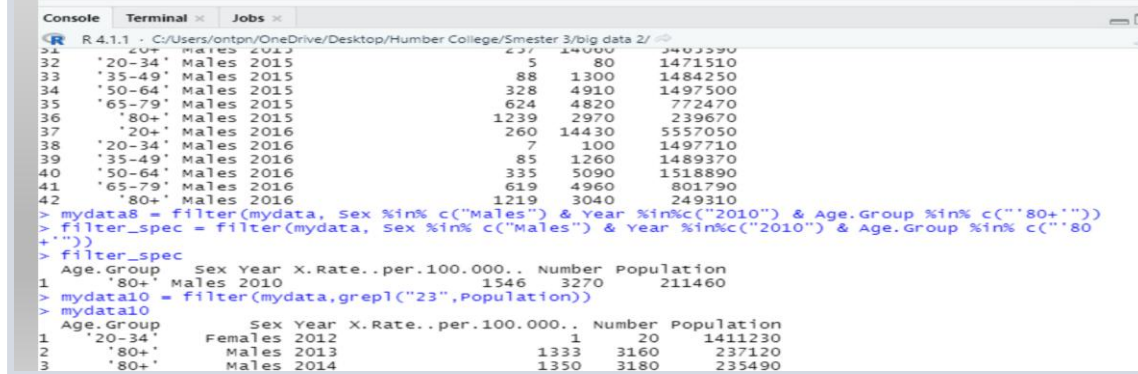
Every sex × year combination from 2010 to 2016 is present exactly once, so no year or group has gaps.

Filtering and validating

In R

Pulling out exactly the rows a question needs, then re-checking the rates

```
35 mydata10 = filter(mydata, Sex == "Males")
36 # using Pipe and filter getting data of specific column
37 filter_spec = filter(mydata, Sex %in% c("Males") & Year %in% c("2010") & Age.Group %in% c("'80+"))
38 filter_spec
39 # using Filter and grepl function to show value in population column having figure 23
40 mydata10 = filter(mydata, grepl("23", Population))
41
42
```

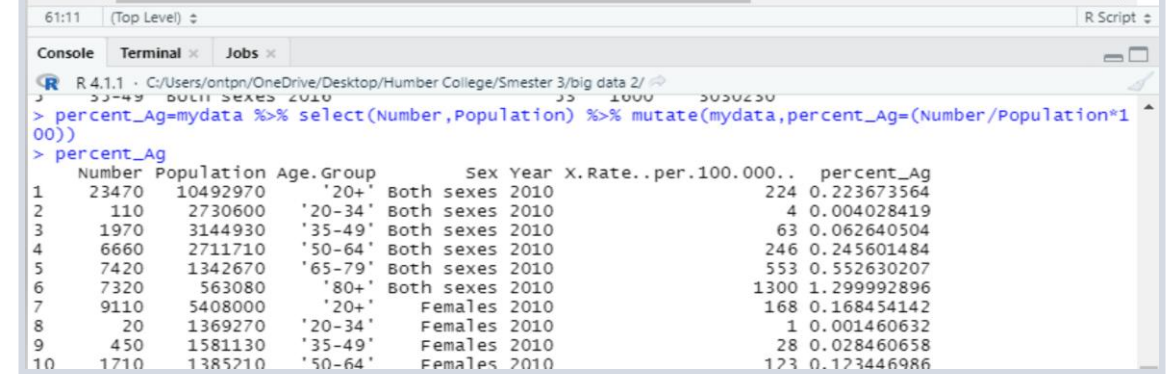


Age.Group	Sex	Year	X.Rate..per.100.000.	Number	Population
'80+'	Males	2010	1546	3270	211460

filter() — ask a precise question

Example: men, aged 80+, in 2010 returns exactly one row (rate 1,546 per 100,000). Every chart starts with a filter like this.

```
59 # percentage of affected people from heart disease by adding new variable
60 percent_Ag=mydata %>% select(Number,Population) %>% mutate(percent_Ag=(Number/Population*100))
61 percent_Ag
```



Number	Population	Age.Group	Sex	Year	X.Rate..per.100.000.	percent_Ag
23470	10492970	'20+'	Both sexes	2010	224	0.223673564
110	2730600	'20-34'	Both sexes	2010	4	0.004028419
1970	3144930	'35-49'	Both sexes	2010	63	0.062640504
6660	2711710	'50-64'	Both sexes	2010	246	0.245601484
7420	1342670	'65-79'	Both sexes	2010	553	0.552630207
7320	563080	'80+'	Both sexes	2010	1300	1.299992896
9110	5408000	'20+'	Females	2010	168	0.168454142
20	1369270	'20-34'	Females	2010	1	0.001460632
450	1581130	'35-49'	Females	2010	28	0.028460658
1710	1385210	'50-64'	Females	2010	123	0.123446986

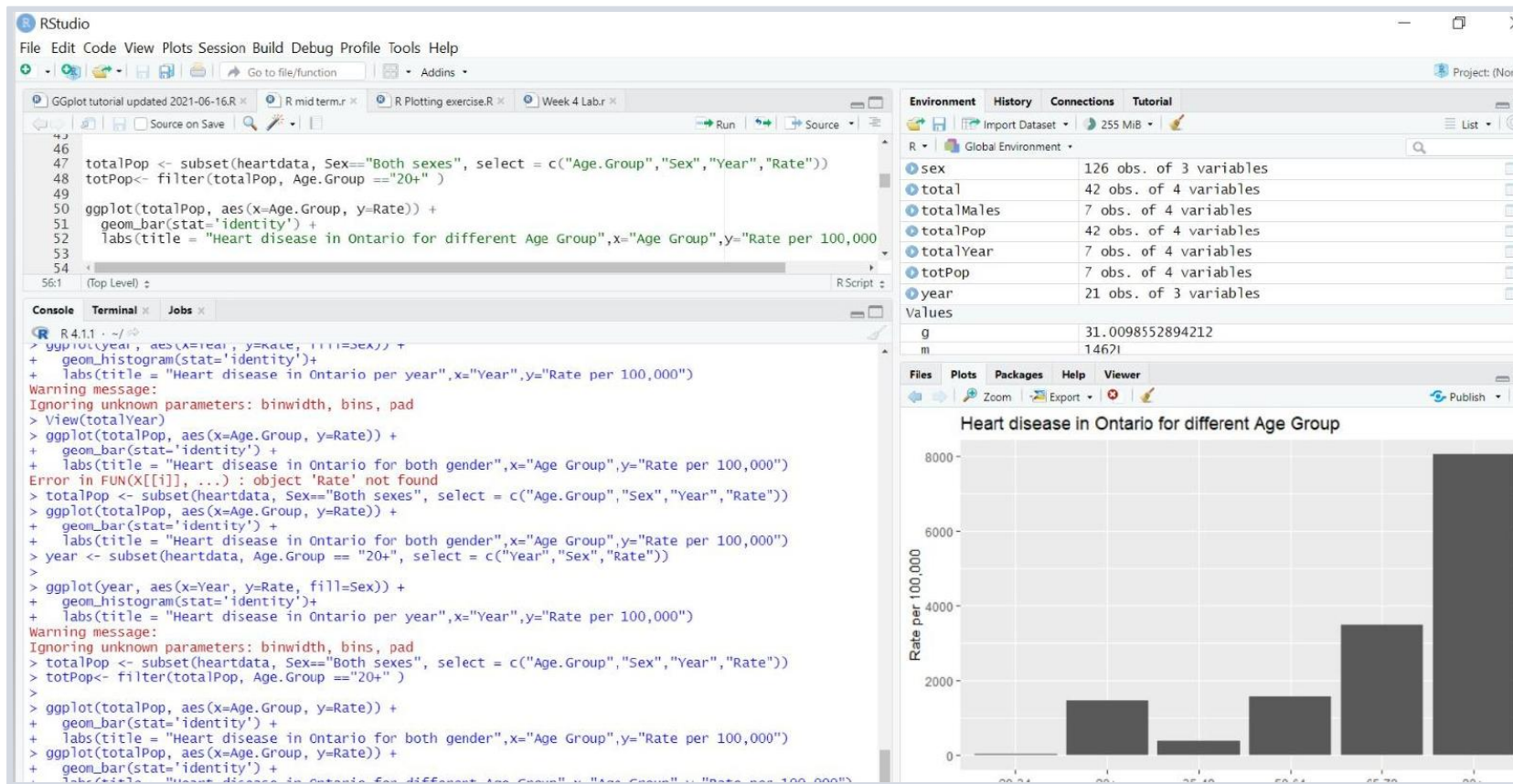
mutate() — add a check column

Cases ÷ Population gives the % affected. $23,470 \div 10,492,970 = 0.224\%$, i.e. 224 per 100,000 — matching the published rate.

From code to chart

In R

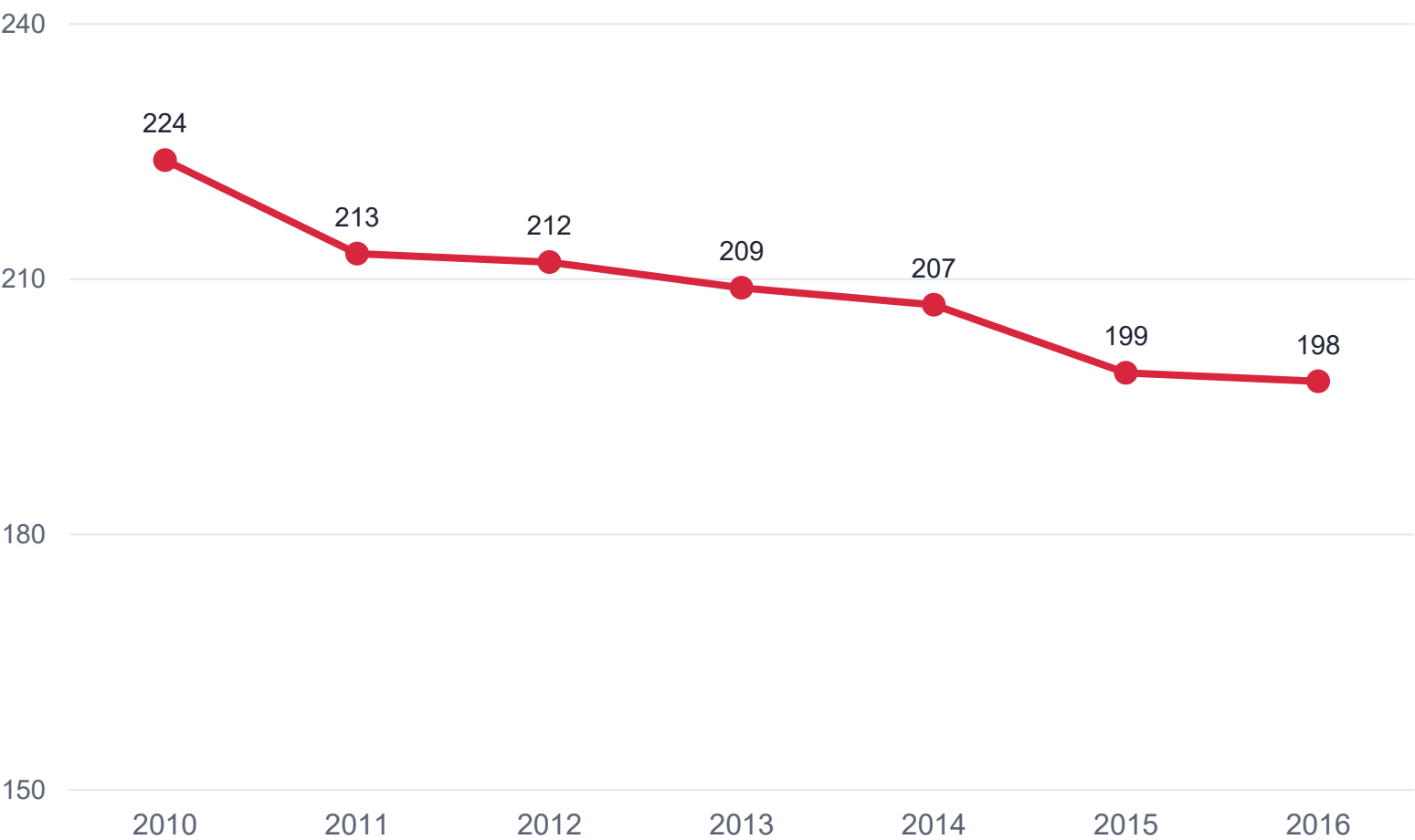
A few lines of ggplot2 turn a filtered table into a graph



- 1 **subset()**
keep "Both sexes" rows
- 2 **filter()**
keep one age group
- 3 **ggplot()**
map age to x, rate to y
- 4 **geom_bar()**
draw the bars

Finding 1 — The adult rate fell about 12%

Heart disease rate per 100,000 adults (20+, both sexes)



2010 and 2016 values are from the dataset; 2012–2015 read from our R chart.

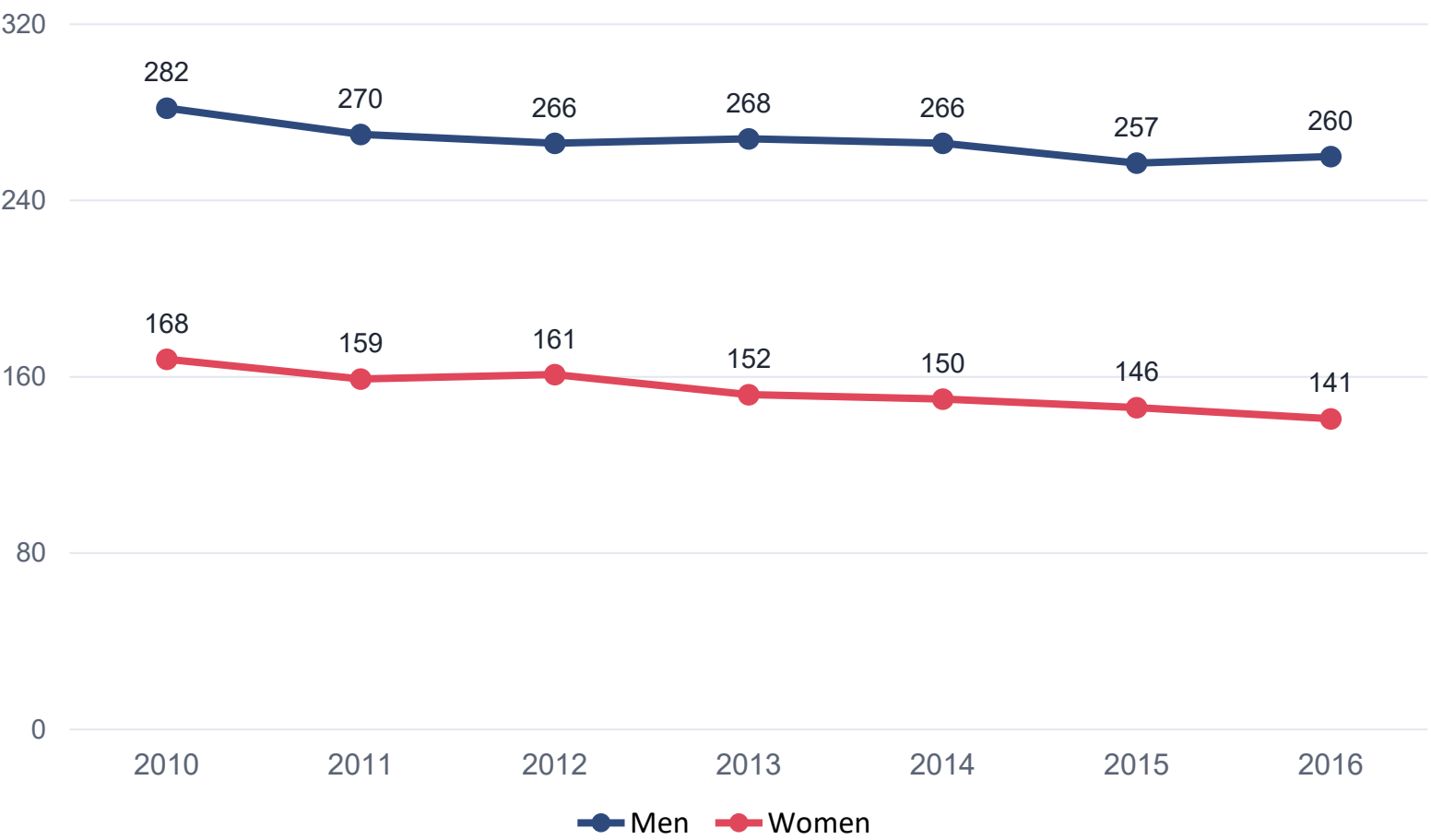
-12%

224 → 198 per 100,000

A steady decline every single year.
The biggest drops were in older adults: 65–79 fell 18% and 80+ fell 22%.

Finding 2 — Men’s rate is about 1.7× women’s

Heart disease rate per 100,000 adults (20+), by sex



1.7×

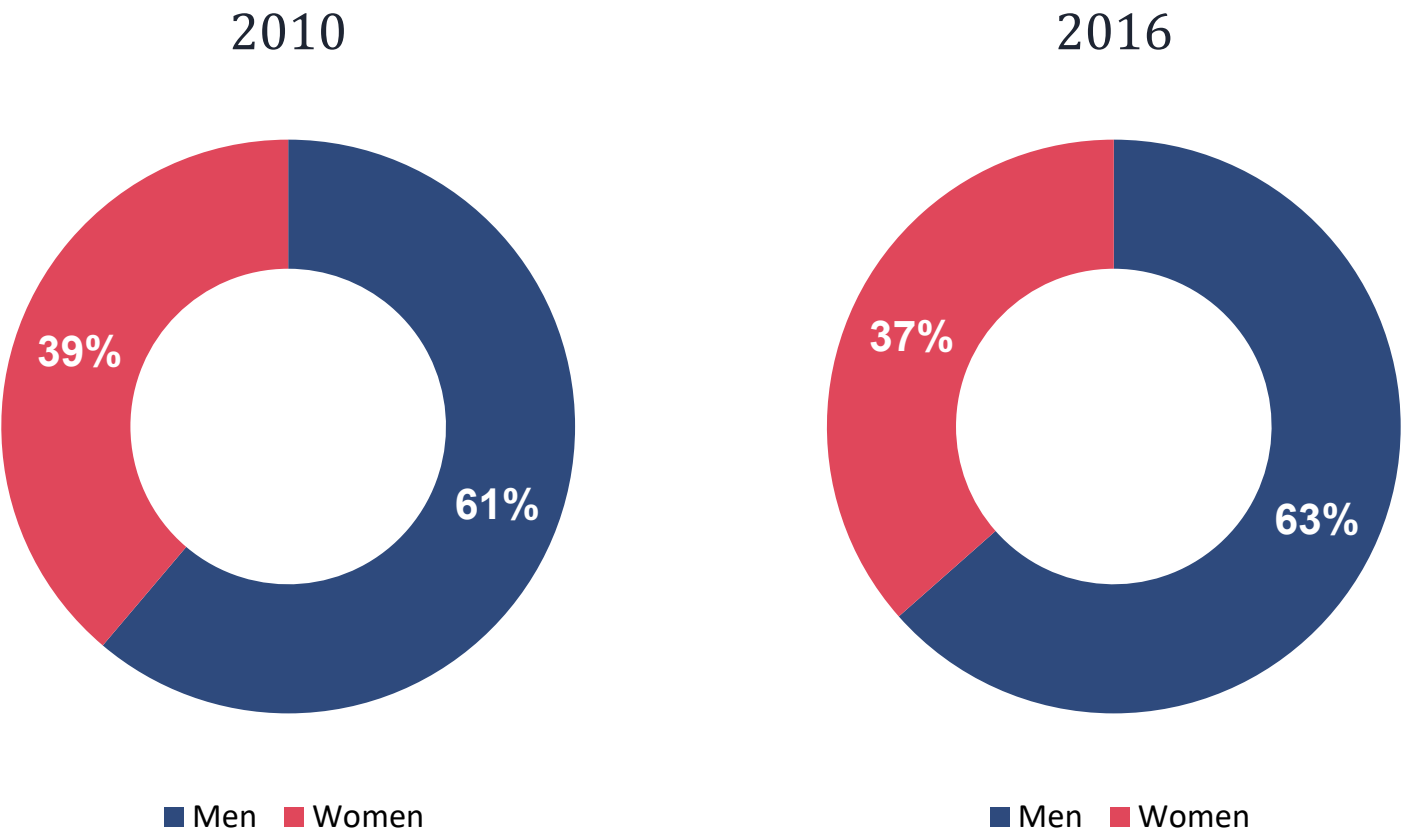
men vs women, every year

Both rates fell, but the gap did not close (1.7× in 2010, 1.8× in 2016). The gap is widest in middle age: at 35–64, men’s rate is 3–4× women’s.

2010 and 2016 values are from the dataset; 2011–2015 read from our R chart.

Men make up about 6 in 10 heart disease cases

Share of adult (20+) heart disease cases by sex



61% → 63%

men's share of cases

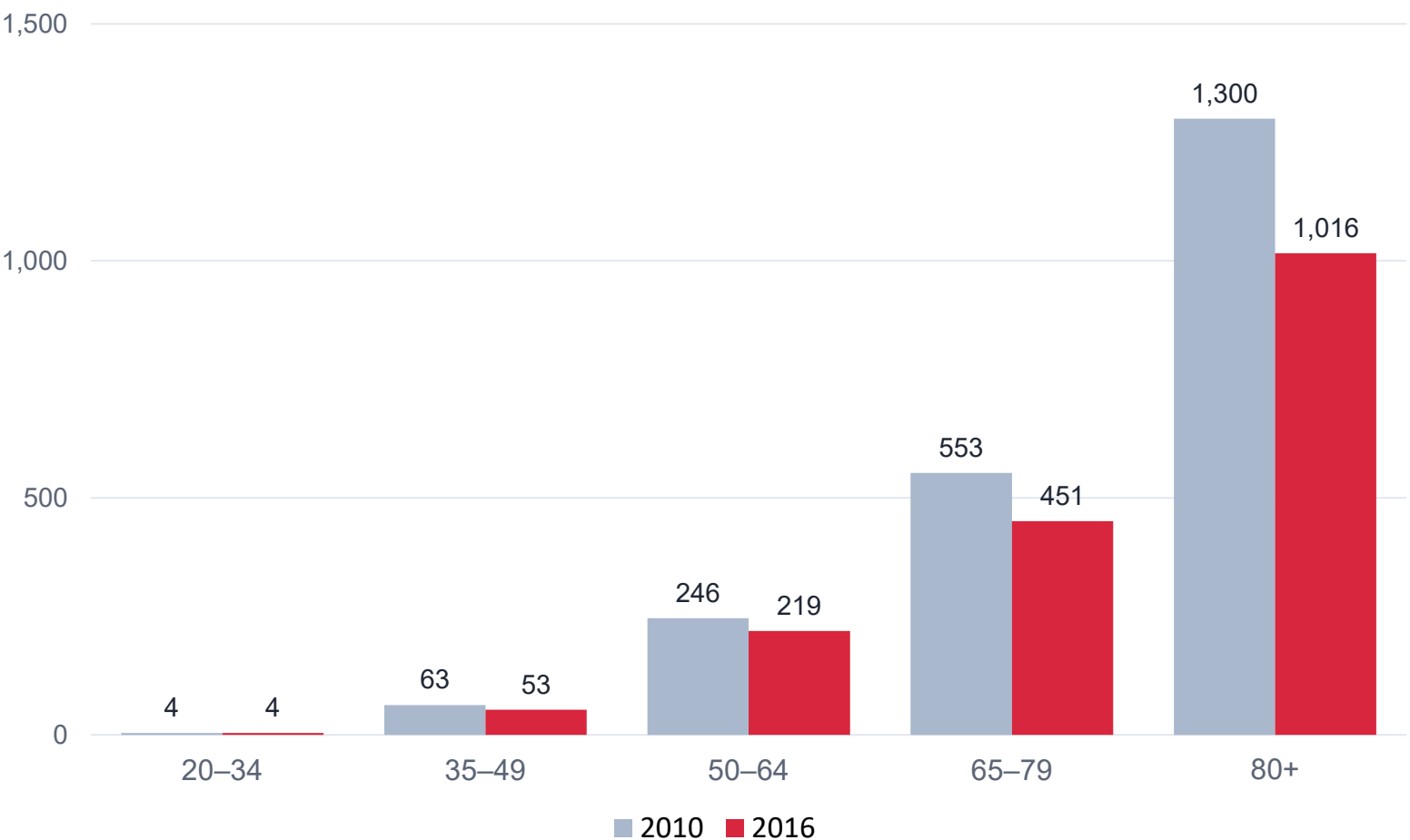
2010: 14,360 men vs 9,110 women.
2016: 14,430 men vs 8,300 women.

Women's cases fell faster, so men's share grew slightly.

Men and women only. The "Both sexes" rows are totals, so they are left out to avoid counting everyone twice.

Finding 3 — Risk climbs steeply with age

Rate per 100,000, both sexes, 2010 vs 2016



~300×

higher rate at 80+ than at 20-34

Rates roughly double with each 15-year band after 35. Yet the oldest groups also improved the most: 80+ went from 1,300 to 1,016.

Finding 4 — Rates fell, but cases barely did

Why? Ontario's adult population grew and got older, 2010 → 2016

+9%

adult population

10.49M → 11.45M

-3%

heart disease cases

23,470 → 22,730

-12%

rate per 100,000

224 → 198



The 65–79 population grew 29% and 80+ grew 13% — the groups with the highest risk.

Adults 65+ account for about 63% of all cases in both 2010 and 2016.

So individuals are at lower risk than before, but hospitals still see nearly as many patients.

What cleaning changed in our results

Three conclusions from our first draft that did not survive checking



“Women have more heart disease”

The first comparison summed the population column, not cases. Using cases: men 14,360 vs women 9,110 in 2010 — men are ~61% of cases.



Pie chart: “Both sexes” = 50%

The pie added the “Both sexes” total to men and women, counting everyone twice. That slice will always be exactly half.



“Young adults (20+) are at high risk”

“20+” is all adults combined, not a young group. The real youngest band, 20–34, has the lowest rate: 4 per 100,000.

Takeaways



Improving

Adult heart disease rate fell 12% from 2010 to 2016, every year.



Men at higher risk

About 1.7× women's rate and ~6 in 10 cases; 3–4× in middle age.



Age drives risk

80+ rate is ~300× that of 20–34; older groups improved most.



Demand stays high

An older, larger population keeps case numbers nearly flat.

Next step for OHMA: focus awareness on men 35–64, where the gap with women is widest and prevention has the most years to pay off.



Data source

Public Health Ontario — heart disease cases, rates per 100,000 and population, Ontario adults 20+, 2010–2016



Explore the project

github.com/n01468012

Thank you — questions?